MIL-S-7780B(WEP)
AMENDMENT 3
2 May 1960
Superseding
Amendment 2
25 June 1958

MILITARY SPECIFICATION

STARTERS: ELECTRIC, CONSTANT CURRENT, DC

This amendment forms a part of Military Specification MIL-S-7780B(AER), 30 March 1956, and has been approved by the Bureau of Naval Weapons, Department of the Navy.

By this Amendment basic Specification MIL-S-7780B(Aer) is changed to read MIL-S-7780B(Wep).

Page 1, paragraph 1.1: Revise to read as follows:

1.1 Scope.- This specification covers electric variable voltage direct current starters for operation from a constant current power source and constant voltage starters for operation from an aircraft battery or a current limited power source.

Page 1, paragraph 1.2 Classification: Add:

Type E-5 starters operating from a constant voltage current limited power source having a rated output drive speed of 6000 RPM and a rated output torque of 7.5 foot pounds.

Page 1, paragraph 2.1: Add:

MIL-B-8565(AER)	Batteries, Storage Aircraft, Maintenance Free, General Specification for
MIL-M-18872(AER)	Microfilming Engineering Drawings and Related Data, Requirements for

Page 2, paragraph 2.1 Air Force-Navy Aeronautical Standard Drawing: Add:

AND20009	Drive, Type XIX, Engine Accessory
AND10264	Flange, Type XIV, Accessory Mounting

Page 2, paragraph 3.3: Revise to read as follows:

3.3 Design and construction.- The starters shall be of compact design and of such construction as to withstand vibration and other severe usage encountered under service conditions. The starters shall be capable of being operated throughout a temperature range of -55° to +70°C (-67° to

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+158°F), and shall not be damaged when exposed to temperatures from -75° to +135°C (-103° to +275°F).

Page 3, Add the following paragraph:

3.3.1.1 Altitude.- The Type E-5 starter shall function and operate satisfactorily at all altitudes up to 30,000 feet.

Page 3, paragraph 3.3.3.1: Revise as follows:

3.3.3.1 Quick detachable mounting.- Type E-3 and Type E-4 starters shall be provided with a quick detachable mounting of a design approved by the procuring activity. This mounting shall not require special tools for its use. It shall be automatically locked when on the engine, and there shall not be any unattached parts, such as clamp rings, bolts, or nuts when the starter is removed from the engine. Any adapter required for modifying the engine accessory drive in order to mount the starter shall be furnished with the starter, but it shall be considered as permanently attached to the engine after initial installation. The design of the quick-detachable mounting shall be such as to permit installation and removal of the starter when there is access to the starter at any one position. Type E-5 starters shall not incorporate quick-detachable mounting features.

Page 3, paragraph 3.3.3.2: Add following sentence:

Type E-5 starters shall be suitable for mounting on Engine Accessory Drive Drawing AND20009, Type XIX except for the changes required to conform with the dimensions shown in figure 2. Type E-5 starters shall provide a drain boss in the starter flange in accordance with Air Force-Navy Aeronautical Standard Drawing "AND10264 Type XIV Accessory Mounting".

Page 3, paragraph 3.3.3.3: Add the following sentence:

Type E-5 starters shall comply with the requirements of this paragraph provided the voltage applied at the starter terminals is not less than 10 volts.

Page 3, paragraph 3.3.4: Add the following sentence:

Type E-5 starters shall have a counterclockwise drive rotation as viewed from the anti-drive end of the starter.

Page 3, paragraph 3.4: In the first sentence delete the words "variable voltage".

Page 3, paragraph 3.4.1: Revise first sentence to read as follows:

Type E-3 and Type E-4 starters shall be provided with an ungrounded motor.

Page 3, Add the following paragraph:

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3.4.1.1 Motor.- Type E-5 starters shall be provided with an ungrounded motor. Electrical connections to the starter shall be made to two insulated terminal pins, located as shown on figure 2. Type E-5 starters shall be designed to operate from either of the following power supplies:

(a) Battery supply.- The battery shall be one 34 ampere-hour, 24 volt nickel-cadmium type conforming to Specification MIL-B-8565(AER).

(b) Ground power supply.- The ground power unit shall supply constant voltage power which will be limited in output current to 650 plus 100, minus zero amperes until the voltage at the generator terminals of the ground power unit is 28 volts, plus zero, minus one volt at which time the current begins to reduce to 250-200 amperes where the cranking cycle is cutoff by an undercurrent relay.

Page 3, paragraph 3.4.3: Revise to read as follows:

3.4.3 Size.- The size of type E-3 and type E-4 starters shall not exceed the overall dimensions shown on figure 1. The size of type E-5 starters shall not exceed the overall dimensions shown on figure 2, and shall provide the cowling mounting bracket as shown on figure 2. The overhung moment of type E-5 starters shall not exceed 80 inch-pounds.

Page 4, paragraph 3.4.4: Add:

Type E-5 Starter: 16.5 pounds.

Page 4, Add paragraph 3.4.5:

3.4.5 Brush inspection access.- Type E-5 starters shall provide an opening and cover for brush inspection and replacement. The cover for the opening shall fit snugly to protect the motor. The use of felt or similar material is prohibited. The brushes shall be accessible for replacement when brush opening covers are removed. The brushes shall be marked by a suitable method, to indicate 75 per cent of the usable length of brush. The design of the brush and/or holder shall be such that the motor armature shall not be subject to mechanical damage if the brush is not replaced when the wear exceeds the allowable usable length.

Page 6, Add paragraph 3.11.1.1:

3.11.1.1 Nameplate.- Type E-5 starter nameplate shall conform to paragraph 3.11.1 herein except as follows:

STARTER: Engine, Electric, DC

Page 8, paragraph 4.4: Add the following to the end of the first sentence:

"or paragraph 3.4.1.1 whichever is applicable".

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Page 8, paragraph 4.5.2: Revise to read as follows:

4.5.2 Torque-speed output.- The starter shall be tested to determine conformance with the torque-speed output requirements of Table I at the input current specified.

TABLE I

Torque-Speed Output

Type	Volts +0,-1	Jaw Speed RPM (Min)	Continuous Torque lb-ft	No-load Speed RPM (Min)	Max Allowable Torque lb-ft	Current Amperes (Max)
E-3	30	850	135±2	1600	450	1000
E-4	30	2000	61±1	3500	450	1000
E-5	24	9500	3.1±$^{2}_{1}$	12000	25	300
E-5	24	6000	7.5±0.5	12000	25	650

Page 8, paragraph 4.5.3: In the first sentence revise "a current input of 1000 amperes." to read, "the current and voltage characteristics as specified in paragraphs 3.4.1 and/or 3.4.1.1 herein.

Page 8, paragraph 4.5.4: Revise to read as follows:

4.5.4 Continuous Torque.- The starter shall be checked to determine that the output speed is not less than that specified in Table I with the starter terminal voltage and input current as specified in Table I, and an applied continuous torque within the range specified in Table I.

Page 8, paragraph 4.5.5: Revise to read as follows:

4.5.5 Efficiency.- The overall efficiency of the starter at rated torque shall not be less than 53% at the input voltage specified in Table I herein.

Page 8, paragraph 4.5.6: Add Type E-5:

Type E-5 Flywheel Polar Moment of Inertia - 2.4 lb.-ft.^{2}
Phase A - 6000 RPM (Min) (Within 8 seconds)
Phase B - 6000 RPM (Min)
7.5 lb-ft ±0.5 Torque (For a period of 10 seconds)
Phase C 12,000 (within 20 seconds)
Terminal Voltage - 24 volts (Max)

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Page 8, Add the following paragraph:

4.5.6 Endurance.- The starter shall complete the 600-cycle endurance test shown in table II without replacement of, or damage to, any of the component parts. Each cycle shall consist of three consecutive phases as shown in table II. Phase A shall consist of the specified inertia load. After the speed specified in phase A is reached the torque load of phase B shall be added for 10 seconds. This torque load shall then be removed for phase C, which shall be completed in a maximum of 12 seconds. Total brush wear shall not have exceeded 75 percent of the original effective brush length of any one brush. At the end of this test, the starter efficiency shall not have decreased more than 5 percent from the original value. Each cycle may be followed by a rest period of up to 5-minutes duration. Cooling air may be supplied to prevent excessive temperature.

Page 9, paragraph 4.5.7: Revise first sentence to read as follows:

4.5.7 Overspeed.- The starter shall be subjected to a no-load test for a period of 60 seconds with 35.0 volts dc applied at type E-3 and E-4 starter terminals and 28 volts dc applied at type E-5 starter terminals.

Page 9, paragraph 4.5.8.1: In the last sentence delete the following:

"with a current input of 1000 amperes where applicable."

Page 9, paragraph 4.5.8.2: In the last sentence delete the following:

"with a current input of 1000 amperes where applicable."

Page 9, Add paragraph 4.5.10 as follows:

4.5.10 Vibration.- If an engine is available, the starter shall be installed and operated in conjunction with the test of the engine for a minimum of 100 hours of engine operation.

Page 10, paragraph 5.4: Revise to read as follows:

5.4 Marking.- Each package and exterior shipping container shall be marked to comply with Specification MIL-STD-129 and to include the following:

Type E-3 and E-4:
STARTER; ELECTRIC, CONSTANT CURRENT, DC
Specification MIL-S-7780B(WEP)
Type
Date of Manufacture

Type E-5:
STARTER; ENGINE, ELECTRIC, DC
Specification MIL-S-7780B(WEP)
Type
Date of Manufacture

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Page 10, paragraph 6.2.1.1 Drawing Requirements: Revise Specification MIL-D-5028 to read Specification MIL-D-70327.

Page 10, paragraphs 6.2.1.1.1 and 6.2.1.1.2: Delete.

Page 10, paragraph 6.2.1.1.3: Renumber and revise to read as follows:

6.2.1.1.1 Microfilm.- One complete set of detail, subassembly, assembly and installation drawings on microfilm prepared and packaged in accordance with Specification MIL-M-18872(AER) shall be sent to the following destination:

Commanding Officer, Naval Aviation Supply Depot
Philadelphia, Pennsylvania
Attention: Code CDD

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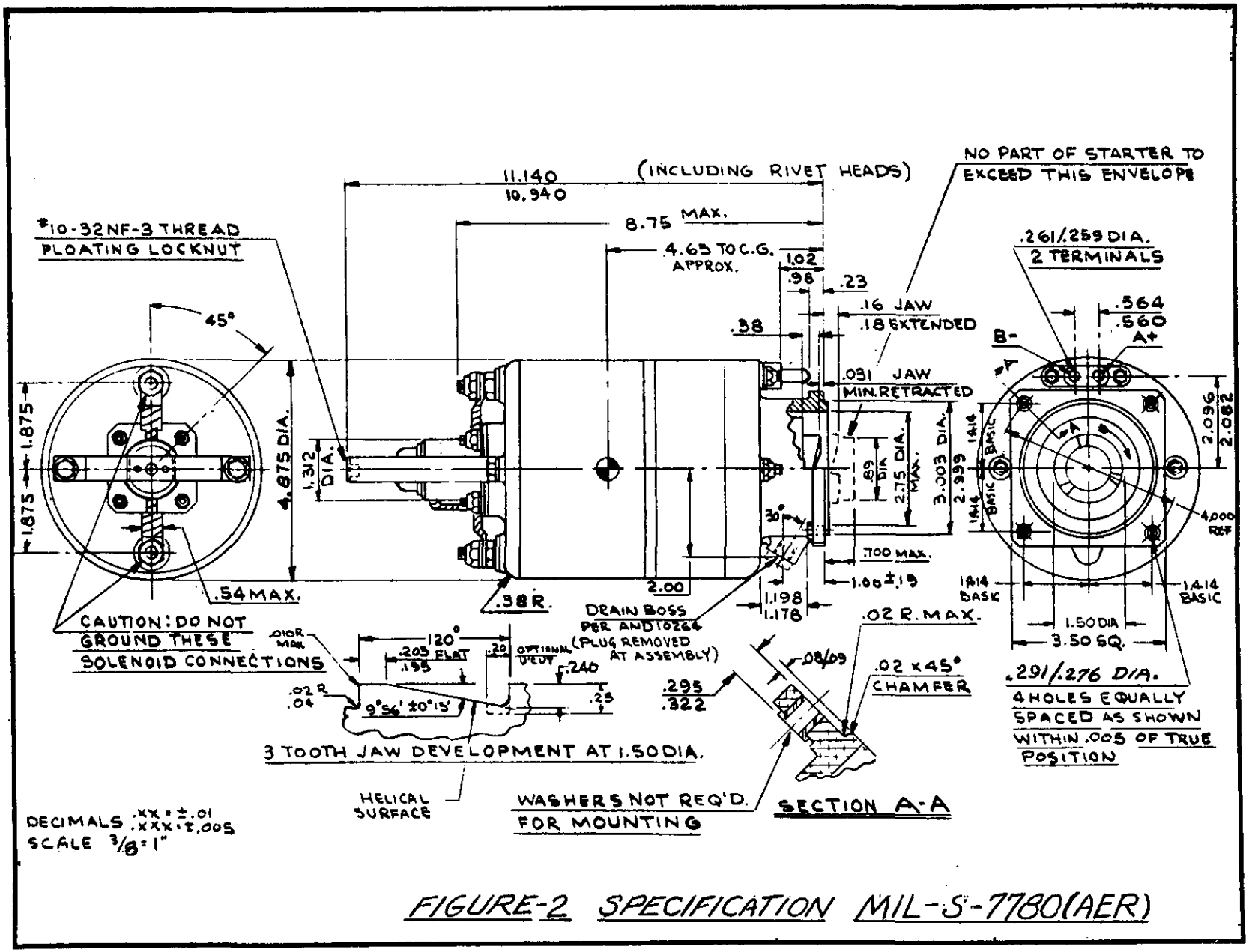


FIGURE-2 SPECIFICATION MIL-S-7780(AER)

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